MTH 207 Practice Exam 3

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating "does not exist" without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

Like any of the arc trig functions:
1. Apply inverse
2. $\frac{d}{dx}$
3. Solve for $\frac{dy}{dx}$
4. Replace y

1. Starting from $y = \log_a(x)$ find a formula for $\frac{dy}{dx}$, showing all steps.

$a^y = x$

$a^y \ln(a) \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{a^y \ln(a)}$

$\frac{dy}{dx} = \frac{1}{x \ln(a)}$

2. Find the linearization of $f(x) = 4^x$ at $x = 3$

$f'(x) = 4^x \ln(4)$

$f'(3) = 4^3 \ln(4)$ $\quad f(3) = 4^3$

point $(3, 4^3)$

$y - 4^3 = 4^3 \ln(4)(x - 3)$ ← This is equation of tangent line

This spot must have $f'(3)$ not $f'(x)$

$y = 4^3 \ln(4)(x - 3) + 4^3$

$L(x) = 4^3 \ln(4)(x - 3) + 4^3$ Not required to simplify 4^3 as 64

3. If $f(2) = 3$ and $f'(x) \leq 7$ for all x, how large can $f(10)$ possibly be?

f is differentiable for all x, so f is continuous for all x.

Must state by name

f is continuous on $[2, 10]$ and differentiable on $(2, 10)$ so by the Mean Value Theorem there is an x-value c in $(2, 10)$ such that $f'(c) = \frac{f(10) - f(2)}{10 - 2}$

$\frac{f(10) - 3}{8} = f'(c) \leq 7$

$\frac{f(10) - 3}{8} \leq 7$

$f(10) - 3 \leq 56$

$f(10) \leq 59$

4. If $y = 2\ln(x) + 3\ln(x^4) + 5\log_6(x) + 7^x - 8x^{10}$ find $\frac{dy}{dx}$

$\frac{dy}{dx} = 2\cdot\frac{1}{x} + 3\cdot\frac{1}{x^4}\cdot 4x^3 + 5\cdot\frac{1}{x\ln 6} + 7^x\ln 7 - 8\cdot 10x^9$

Alternate answer:

$y = 2\ln x + 12\ln x + 5\log_6 x + 7^x - 8x^{10}$

$\frac{dy}{dx} = 2\cdot\frac{1}{x} + 12\cdot\frac{1}{x} + 5\cdot\frac{1}{x\ln 6} + 7^x\ln 7 - 80x^9$

5. If $y = 2^x\cdot\sin(x^3+x^2) + 5\cos x + 6\sec x + 7\tan x + 8\arctan x + 9\arcsin x + \frac{x^3\sin x}{e^x + x^6}$ find $\frac{dy}{dx}$

$\frac{dy}{dx} = 2^x\cdot\cos(x^3+x^2)\cdot(3x^2+2x) + \sin(x^3+x^2)\cdot 2^x\ln 2 - 5\sin x + 6\sec x\tan x + 7\sec^2 x + 8\cdot\frac{1}{x^2+1} + 9\cdot\frac{1}{\sqrt{1-x^2}} + \frac{(e^x+x^6)(x^3\cos x + \sin x\cdot 3x^2) - x^3\sin x(e^x+6x^5)}{(e^x+x^6)^2}$

(parens: $(3x^2+2x)$; parens: (e^x+x^6); parens: $(x^3\cos x + \sin x\cdot 3x^2)$; parens: (e^x+6x^5))

If the variable is in both the base and the exponent, the only way to differentiate is logarithmic differentiation

6. If $y = \sqrt{x^4-x}^{\sin x+\cos x}$ find $\frac{dy}{dx}$

$\ln y = \ln\left(\sqrt{x^4-x}^{\sin x+\cos x}\right)$

$\ln y = (\sin x + \cos x)\ln\left((x^4-x)^{1/2}\right)$ (parens: $(\sin x+\cos x)$)

$\frac{1}{y}\cdot\frac{dy}{dx} = (\sin x+\cos x)\cdot\frac{1}{(x^4-x)^{1/2}}\cdot\frac{1}{2}(x^4-x)^{-1/2}\cdot(4x^3-1) + \ln\left((x^4-x)^{1/2}\right)\cdot(\cos x - \sin x)$ (parens: $(4x^3-1)$; parens: $(\cos x-\sin x)$)

$\frac{dy}{dx} = y\left((\sin x+\cos x)\cdot\frac{1}{(x^4-x)^{1/2}}\cdot\frac{1}{2}(x^4-x)^{-1/2}\cdot(4x^3-1) + \ln\left((x^4-x)^{1/2}\right)\cdot(\cos x-\sin x)\right)$

$\frac{dy}{dx} = \sqrt{x^4-x}^{\sin x+\cos x}\left((\sin x+\cos x)\cdot\frac{1}{(x^4-x)^{1/2}}\cdot\frac{1}{2}(x^4-x)^{-1/2}\cdot(4x^3-1) + \ln\left((x^4-x)^{1/2}\right)\cdot(\cos x-\sin x)\right)$

Where is f' undefined?
Where does f' equal 0?

7. Find the critical numbers of $f(x) = \frac{1}{x-3} + \frac{x}{4}$ $= (x-3)^{-1} + \frac{1}{4}x$

$f'(x) = -1(x-3)^{-2} \cdot (1-0) + \frac{1}{4} \cdot 1$

$= \frac{-1}{(x-3)^2} + \frac{1}{4}$ undefined at $x = 3$

$\frac{-1}{(x-3)^2} + \frac{1}{4} = 0$

$\frac{1}{4} = \frac{1}{(x-3)^2}$

$(x-3)^2 = 4$

$x - 3 = \pm 2$

$x = 3 \pm 2$

$x = 1$ or $x = 5$

1, 3, and 5 are the critical numbers of f.

8. Find the absolute max/min of $f(x) = e^{-x^3}$ on $[-1, 1]$

$f'(x) = e^{-x^3} \cdot -3x^2$ always defined.

$e^{-x^3} \cdot -3x^2 = 0$

$\frac{-3x^2}{e^{(x^3)}} = 0$

$-3x^2 = 0$

$x = 0$ ← 0 is the only critical number of f.

Check endpoints

$f(0) = e^{-0^3} = e^0 = 1$

$f(-1) = e^{-(-1)^3} = e^{-(-1)} = e^1 = e$

$f(1) = e^{-(1)^3} = e^{-1} = \frac{1}{e}$

f has abs max e at -1. f has abs min $\frac{1}{e}$ at 1.

9. Find all local max(s)/min(s) of $f(x) = x^3 - 3x^2 - 24x + 207$

Answer based on First Deriv. Test

$f'(x) = 3x^2 - 6x - 24$ always defined

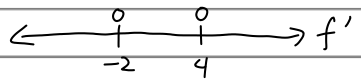
$3x^2 - 6x - 24 = 0$

$3(x^2 - 2x - 8) = 0$

$3(x+2)(x-4) = 0$

$3 = 0$ $x+2 = 0$ $x-4 = 0$

no sol. $x = -2$ $x = 4$

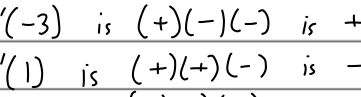


$f'(x) = 3(x+2)(x-4)$

$f'(-3)$ is $(+)(-)(-)$ is $+$

$f'(1)$ is $(+)(+)(-)$ is $-$

$f'(5)$ is $(+)(+)(+)$ is $+$



f has local max at -2 (because f' changes from + to −)

f has local min at 4 (because f' changes from − to +)

Answer based on Second Deriv. Test

$f'(x) = 3x^2 - 6x - 24$ always defined

$3x^2 - 6x - 24 = 0$

$3(x^2 - 2x - 8) = 0$

$3(x+2)(x-4) = 0$

$3 = 0$ $x+2 = 0$ $x-4 = 0$

no sol. $x = -2$ $x = 4$

$f''(x) = 6x - 6$

$f''(-2) = 6(-2) - 6 = -18 < 0$

so f has local max at -2

$f''(4) = 6(4) - 6 > 0$

so f has local min at 4

10. Where is $f(x) = e^x(x^2 - 2x + 1)$ increasing/decreasing?

$f'(x) = e^x(2x-2) + (x^2-2x+1)e^x$ always defined.

$e^x(2x-2) + (x^2-2x+1)e^x = 0$

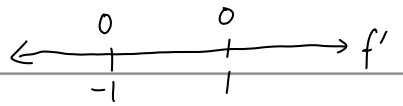
$e^x(x^2+2x-2x-2+1) = 0$

$e^x(x^2-1) = 0$

$e^x(x+1)(x-1) = 0$

$e^x = 0$ $x+1=0$ $x-1=0$

No sol. $x=-1$ $x=1$

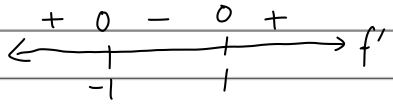


$f'(x) = e^x(x+1)(x-1)$

$f'(-2)$ is $(+)(-)(-)$ is $+$

$f'(0)$ is $(+)(+)(-)$ is $-$

$f'(2)$ is $(+)(+)(+)$ is $+$



f is increasing on $(-\infty,-1) \cup (1,\infty)$

f is decreasing on $(-1,1)$

11. Where is $f(x) = e^x x^4$ concave up / concave down?

$f'(x) = e^x \cdot 4x^3 + x^4 \cdot e^x$

$f''(x) = e^x \cdot 12x^2 + 4x^3 \cdot e^x + x^4 \cdot e^x + e^x \cdot 4x^3$ always defined.

(Where is $f'' = 0$?) $e^x \cdot 12x^2 + 4x^3 \cdot e^x + x^4 \cdot e^x + e^x \cdot 4x^3 = 0$

$e^x(12x^2 + 4x^3 + x^4 + 4x^3) = 0$

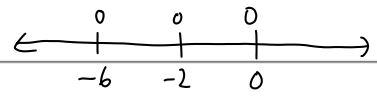
$e^x(x^4 + 8x^3 + 12x^2) = 0$

$e^x \cdot x^2(x^2 + 8x + 12) = 0$

$e^x \cdot x^2 \cdot (x+6)(x+2) = 0$

$e^x = 0$ $x^2 = 0$

No sol $x = 0$ $x = -6$ $x = -2$



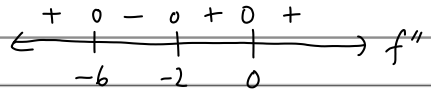
$f''(x) = e^x \cdot x^2 \cdot (x+6) \cdot (x+2)$

$f''(-7)$ is $(+)(+)(-)(-)$ is $+$

$f''(-3)$ is $(+)(+)(+)(-)$ is $-$

$f''(-1)$ is $(+)(+)(+)(+)$ is $+$

$f''(1)$ is $(+)(+)(+)(+)$ is $+$



f is CU on $(-\infty,-6) \cup (-2,0) \cup (0,\infty)$

f is CD on $(-6,-2)$

12. Find all inflection point(s) of $f(x) = x^4 - 6x^2 + 1$

$f'(x) = 4x^3 - 12x$

$f''(x) = 12x^2 - 12$ always defined.

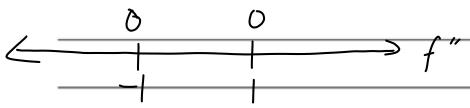
$12x^2 - 12 = 0$

$12(x^2-1) = 0$

$12(x+1)(x-1) = 0$

$12 = 0$ $x+1=0$ $x-1=0$

no sol. $x=-1$ $x=1$

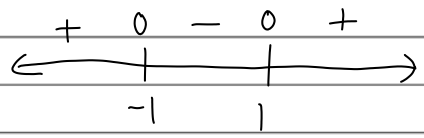


$f''(x) = 12(x+1)(x-1)$

$f''(-2)$ is $(+)(-)(-)$ is $+$

$f''(0)$ is $(+)(+)(-)$ is $-$

$f''(2)$ is $(+)(+)(+)$ is $+$



$f(-1) = (-1)^4 - 6(-1)^2 + 1 = 1 - 6 + 1 = -4$

$(-1,-4)$ is an inflection point

$f(1) = (1)^4 - 6(1)^2 + 1 = 1 - 6 + 1 = -4$

$(1,-4)$ is an inflection point

An inflection point is a point where there is a change in concavity.

$\theta = 7x$

As $x \to 0$ $\theta \to 0$ also

13. Find $\lim_{x\to 0} \frac{\sin(7x)}{5x}$

$$\frac{\sin(7x)}{5x} \cdot \frac{7}{7} = \lim_{x\to 0} \frac{7}{5} \cdot \frac{\sin(7x)}{7x} = \frac{7}{5} \cdot \lim_{x\to 0} \frac{\sin(7x)}{7x} = \frac{7}{5} \cdot \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = \frac{7}{5} \cdot 1 = \frac{7}{5}$$

Recall: $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$

Alternate answer $\lim_{x\to 0} \sin(7x) = \sin(7\cdot 0) = \sin(0) = 0$ $\quad \lim_{x\to 0} 5x = 5\cdot 0 = 0$

From Chain Rule

$$\lim_{x\to 0} \frac{\sin(7x)}{5x} \overset{(l'H)}{=} \lim_{x\to 0} \frac{\cos(7x)\cdot 7}{5\cdot 1} = \frac{7}{5} \lim_{x\to 0} \cos(7x)$$

$$= \frac{7}{5} \cdot \cos(7\cdot 0)$$
$$= \frac{7}{5} \cos(0)$$
$$= \frac{7}{5} \cdot 1$$
$$= \frac{7}{5}$$

14. Find $\lim_{x\to\infty} \frac{\ln x}{x^3}$

$$\lim_{x\to\infty} \frac{\ln x}{x^3} \overset{(l'H)}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{3x^2} = \lim_{x\to\infty} \frac{\frac{1}{x}}{\frac{3x^2}{1}} = \lim_{x\to\infty} \frac{1}{x} \cdot \frac{1}{3x^2} = \lim_{x\to\infty} \frac{1}{3x^3} = 0$$

Typo in version 1: $x \to 0^+$

This should have said $x \to \infty$

Recall if $r > 0$

$$\lim_{x\to\infty} \frac{1}{x^r} = 0$$

15. A malfunctioning pizza dough causes chaos as an ever-growing circle. If the radius is growing at the rate of 3 cm/min, how fast is the area growing when the radius is 11 cm? Include units.



$\frac{dr}{dt} = 3$ $\quad$ Find $\frac{dA}{dt}$ when $r = 11$

↑ Because r is a function of time t, wait to plug in r = 11 until after differentiating.

$$A = \pi r^2$$

$$\frac{dA}{dt} = \pi \cdot 2r \cdot \frac{dr}{dt}$$

$$\frac{dA}{dt} = \pi \cdot 2 \cdot (11) \cdot (3) \ \text{cm}^2/\text{min}$$